Explainability of text embeddings through semantic perturbations and visual analytics

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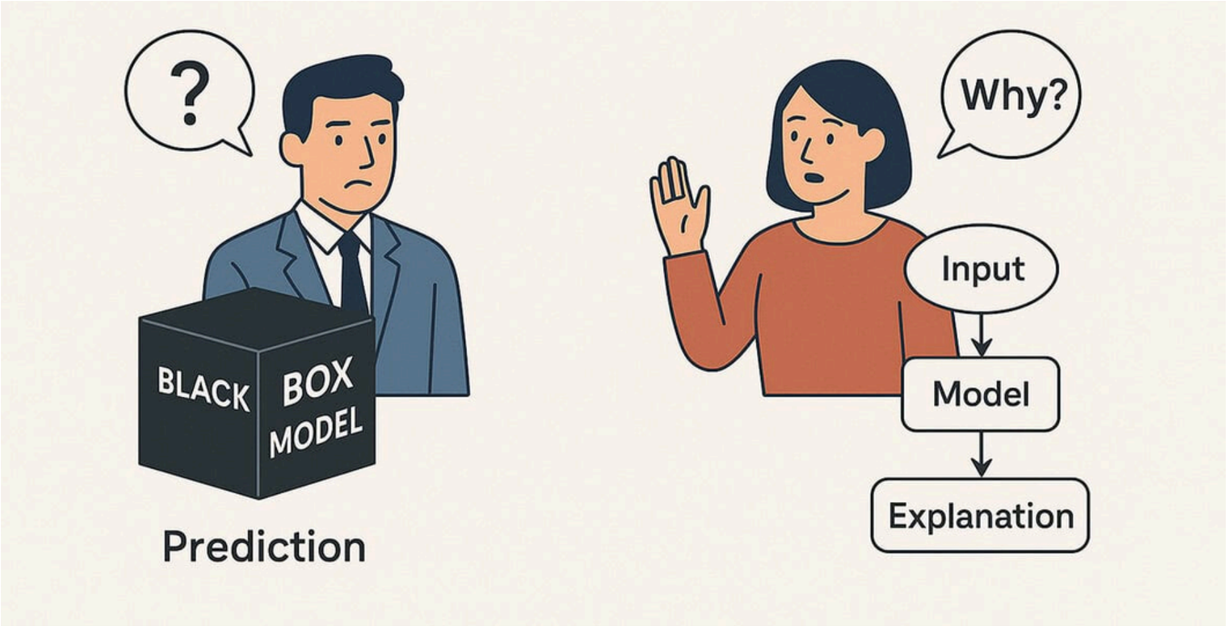
Agenda

- Introduction
- Related Works
- Dataset & Model Selection
- Methodology
- System Architecture
- Explainability Techniques
- Results
- Conclusions



Introduction

Explainable AI (XAI)



Research Question

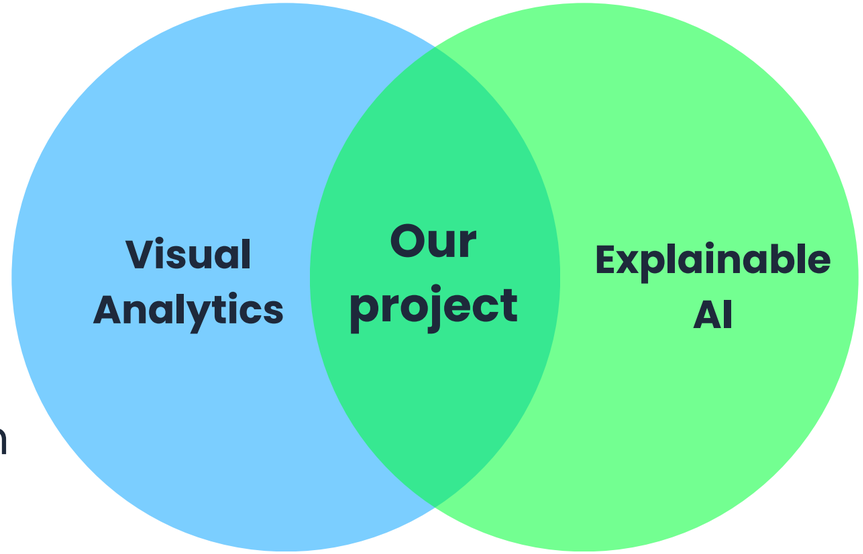
To what extent do syntactic structures and specific words influence the embedding space more than the actual semantic content of the text?

Related Works and Research Gap

Research Intersection

- **Visual Analytics**: Systems like Google's Embedding Projector map global geometries but act as static environments.

- **Explainable AI:**
  - Inspired by LSTMVis, we employ inline text heatmaps to project word-level weights onto the text.
  - CheckList suggested to us the methodology and the system for the perturbations to actively test the model's semantic robustness.



The Research Gap

- The research gap is the absence of a unified human-in-the-loop system connecting spatial exploration , textual analysis and robustness check.

Related Works and Research Gap

Current limitations

Current tools isolate spatial visualization from robustness testing. Visual systems focus on where text is located rather than why it clusters, while traditional robustness tests measure performance without providing visual explanations of the model's behavior

The Shift

- We use BERTopic (c-TF-IDF) to turn raw abstract clusters into interpretable thematic landmarks.
- We analyze embeddings as dynamic vectors. Under controlled linguistic mutations the system calculates the new embedding and instantly visualizes the trajectory of the semantic drift.

Main Contribution

Our main contribution is an interactive tool that tests if the model groups texts based on true semantic meaning or just structural shortcuts.

Dataset & Model Selection

We used the Twitter US Airline Sentiment dataset, composed of real customer-service interactions between users and U.S.A. airline companies from February 2015.

Embedding Model: Qwen3-embedding 0.6B generates the high-dimensional latent space.

Perturbation Model: Gemma executes generative semantic mutations.

Tweet Id	Airline Sentiment	Airline	Text
570295459631263746	positive	Virgin America	@VirginAmerica it was amazing, and arrived an hour early. You're too good to me.
570048726380765184	negative	United	@united I send an email about my bad experience and you send back a generic response. Yet another reason why I'll never fly with you again
569667602148151296	neutral	Southwest	@SouthwestAir how do I get my companion pass?

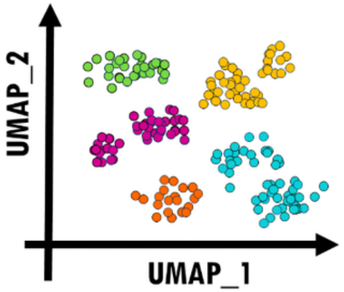
Sample size:

10.445

Methodologies

1. Dimensionality Reduction: UMAP

To make the high-dimensional embedding space visually inspectable, we use **non-linear dimensionality reduction** instead of Linear DR to project the 1024-D tweet vectors into a 2D map.



2. Topic Modeling and Keyword Extraction

Once the projection is created, the system supports cluster interpretation through topic modeling at a global level with **BERTopic** and keyword extraction at a local level through **TF-IDF**.

3. Word Ablation Heatmap

To evaluate the local impact of individual tokens, the system applies a Leave-One-Word-Out (LOWO) routine, measuring the resulting semantic displacement and rendering it as an intuitive, inline text heatmap

4. Targeted Perturbations

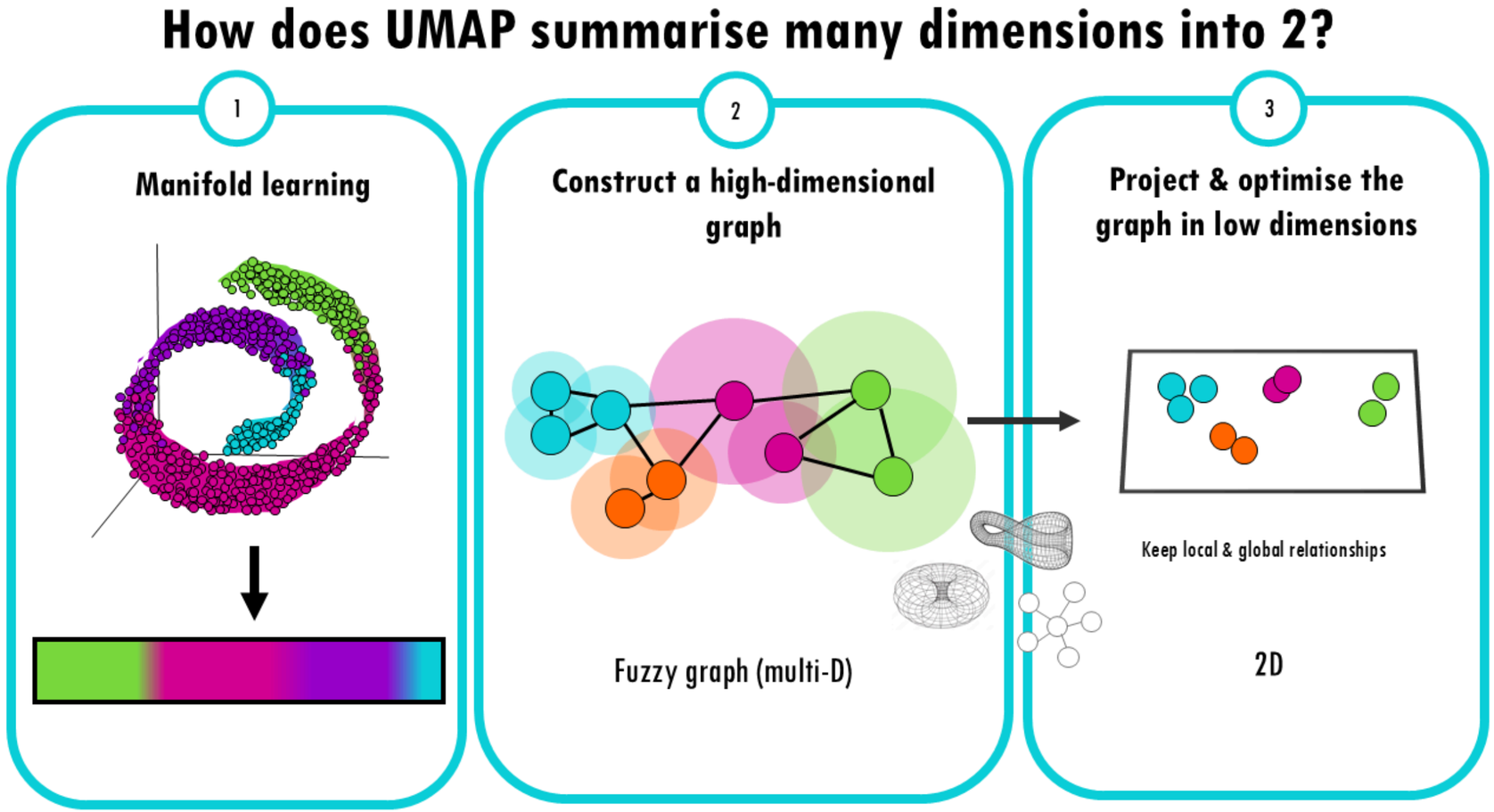
The system supports different perturbation strategies allowing the users to test whether the embedding model is robust to linguistic variations or overly sensitive to superficial textual changes.

WordNet Offline

datamuse

1. UMAP - Uniform Manifold Approximation and Projection

To project our 1024-D Qwen3 embeddings into a 2D space, we implemented UMAP, which optimizes global layout preservation while maintaining local neighborhood relationships.



1. UMAP - Hyperparameters

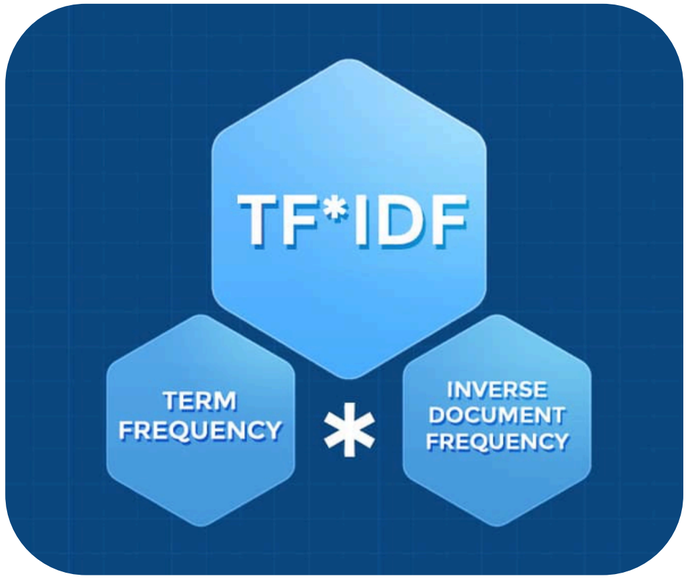
UMAP has several hyperparameters that can have a significant impact on the resulting embedding.

- **n_neighbors = 15**: determines how many nearby points are considered when constructing the high-dimensional graph.
- **min_dist = 0.1**: controls how close points are allowed to become in the final two-dimensional space.
- **metric = 'cosine'**: Cosine distance compares the direction of two embedding vectors rather than their absolute magnitude.
- **random_state = 42:** This guarantees reproducibility, meaning that the same dataset and configuration generate the same two-dimensional coordinates across different executions.

2. TF-IDF and BERTopic

TF-IDF stands for Term Frequency–Inverse Document Frequency. Measures word importance by identifying keywords that appear frequently within a user selected group of tweets, but rarely across the entire dataset.

$$\text{TF-IDF}(t,d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \times \log\left(\frac{N}{df(t)}\right)$$

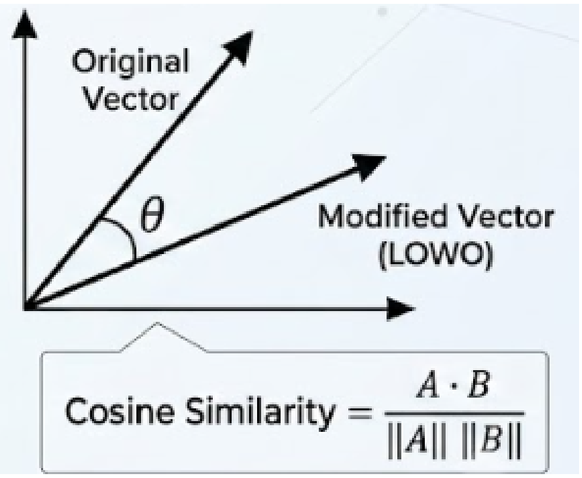


BERTopic is a topic modeling method that identifies latent themes in a collection of texts. It automatically groups semantically similar text vectors into discrete geometric clusters using a combined UMAP and HDBSCAN pipeline.
It implements **c-TF-IDF** merging all tweets in a cluster into a single class-document to evaluate keyword distinctiveness at the topic level rather than individual text level.

3. Word Ablation Heatmap

LOWO Processing: Automates a Leave-One-Word-Out procedure, removing one word at a time to generate a mutated text variant T′ and compute the high-dimensional vector shift.

$$\text{Word Weight} = 1 - \text{Cosine Similarity}(E(T), E(T'))$$



Color Encoding: Maps weights directly onto an interactive inline text spectrum:

- Light Yellow: High semantic impact.
- Dark Blue: Low semantic impact.



We work at the **word level** rather than at token level because complete words are intuitive for users to understand and directly applicable for text perturbations.

4. Targeted Perturbations

Noise-based perturbations

Automated attacks to simulate real-world media noise.

- **Random Noise**
- **Typos**
- **Leetspeak**
- **Shuffling**

Word-based perturbations

Applied to user-selected words (via diagnostic heatmap) or auto-determined by the system.

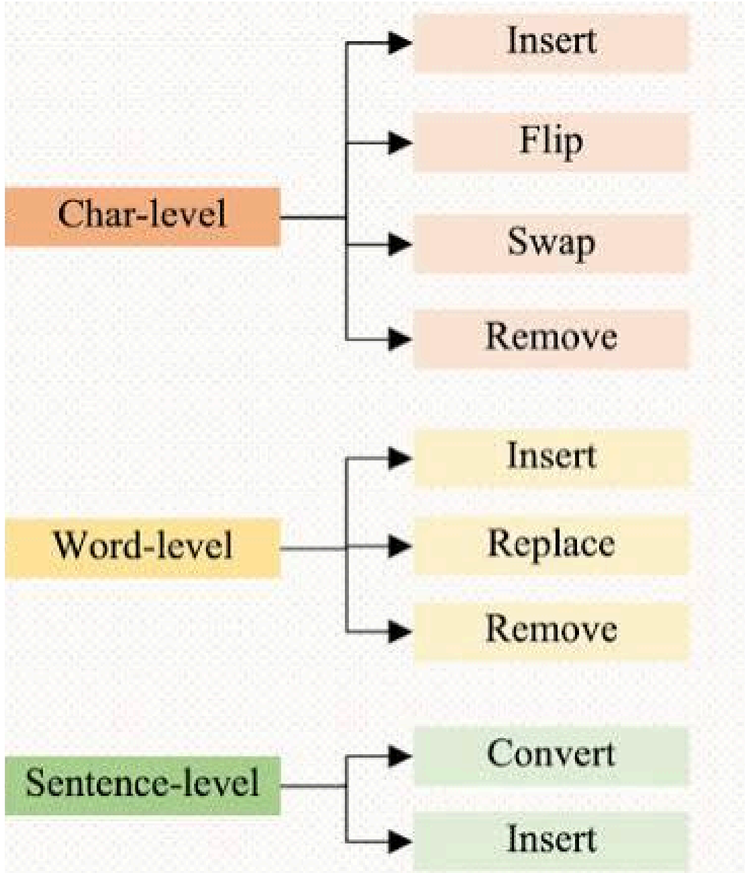
- **Synonym**
- **Antonym**
- **Word Removal**
- **Custom Word Replacement**

LLM-based perturbations

Whole-tweet modifications executed via specialized Gemma model prompting.

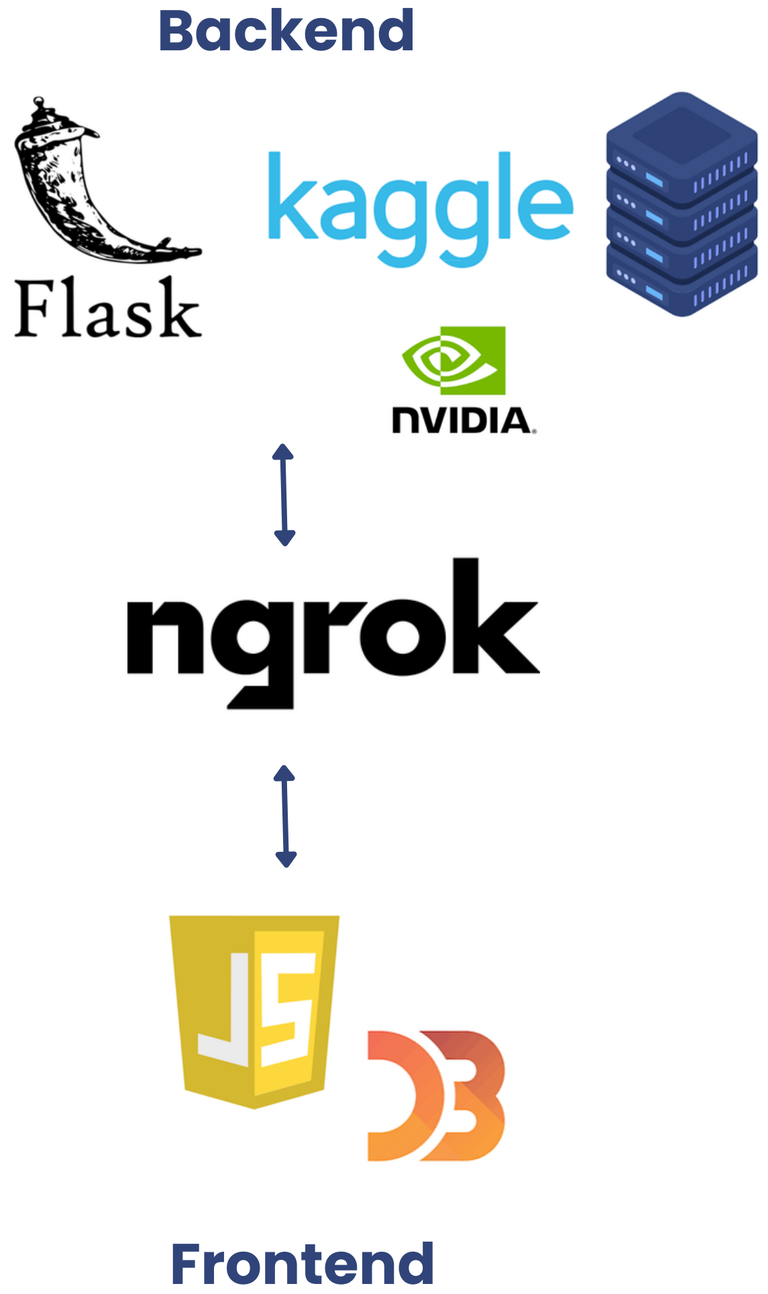
- **Paraphrase**
- **Opposite**

Perturbations-level



System Architecture

- **Cloud GPU Backend:** Executes heavy vector processing in a **Kaggle** environment powered by an **NVIDIA Tesla T4 x 2 GPU** running the Qwen3-embedding:0.6B model.
- **Ngrok** Tunnel Bridge: A Flask API exposes backend functionality, securely bridged to the local browser via an Ngrok secure tunnel
- **Frontend Analytics**: Utilizes JavaScript and D3.js to handle canvas rendering, coordinated dashboard views and real-time trajectory arrow animations.

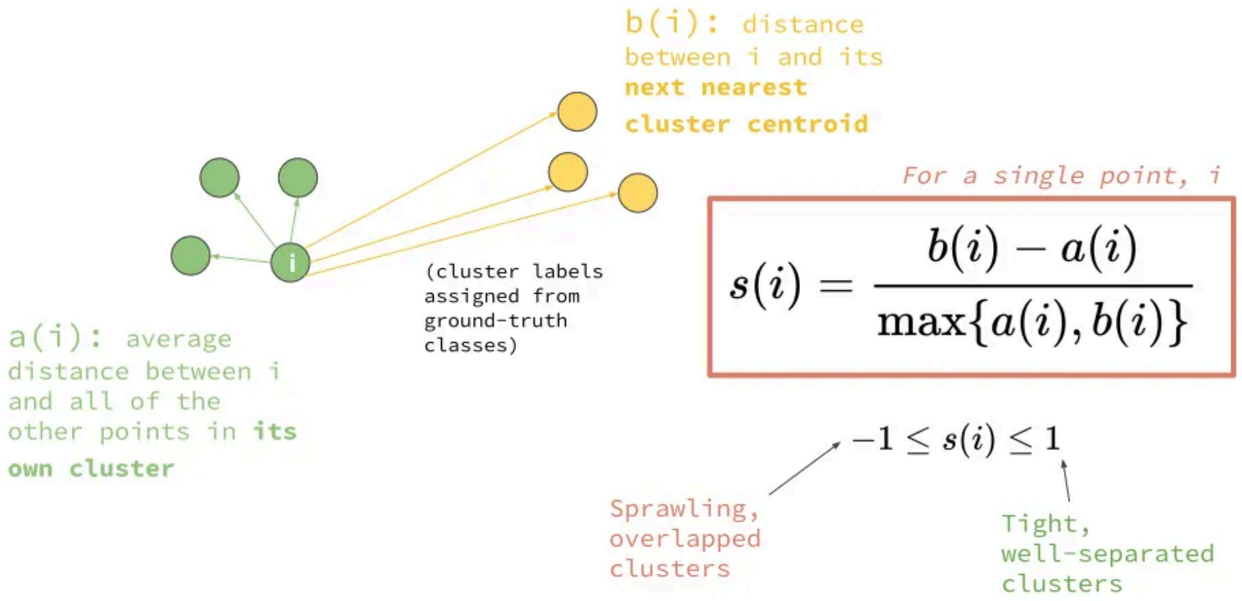


Results I – High Dimensional Space Analysis

What the user can do :

- **Switch the visual grouping**
  We identified three different clusterizations (Airline, Sentiment or Topic) for the user to analyze the same embedding space.

- **Inspect the overlap**
  The user can directly visualize the spatial distance between tweet clusters, enabling the exploration of individual tweets or sub-clusters to extract valuable insights into the underlying semantic divisions.



Main idea: While the automated metric tells us how much clusters overlap, the visual interface reveals where and helps understand why.

Results I – High Dimensional Space Analysis

By switching views, the user discovers that:

- Clusters are not well separated.
- Airline provides the strongest, although still weak, spatial signal.
- Specific overlapping regions can be explained through the analysis of single tweets.

Grouping for:	Silhouette Score
Airline	**0.0755**
Sentiment	**0.0723**
Topic	**0.0101**

Peculiar finding: the exceptionally low inter-cluster distance between US Airways and American Airlines (0.2107) reflects the fact that the dataset was collected in 2015, when the two airlines were merging. During this period, users frequently tagged both airlines due to confusion over responsibility.

Main Insight

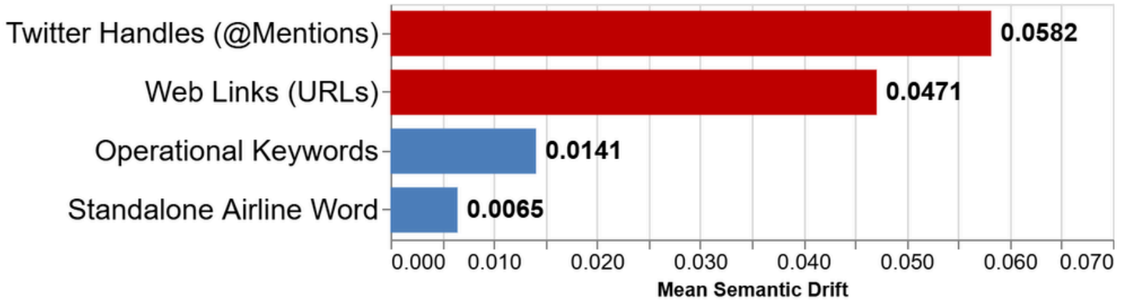
The raw Qwen3 embedding space is organized primarily by **who** is mentioned rather than **what** is discussed

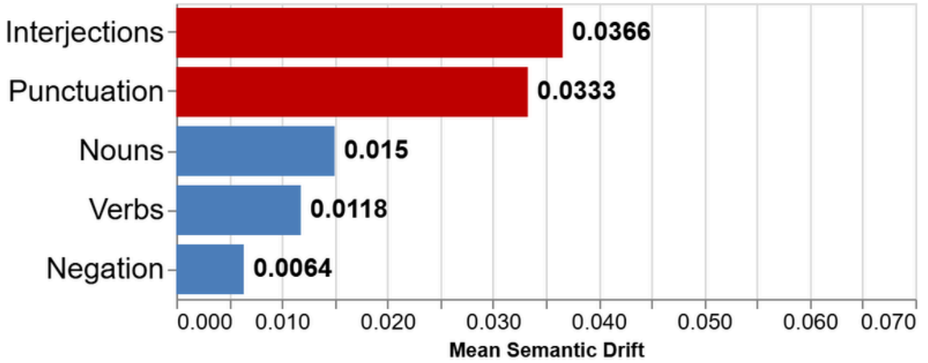
Results II: Local Explainability

LOWO Ablation

We performed a Leave-One-Word-Out ablation on 10,445 tweets to measure which word types most strongly affect the Qwen3 embedding space. Semantic drift was computed after removing one word at a time.

Syntax vs Operational Meaning results:





Main Insight

The Qwen3 embedding space is strongly influenced by Twitter-specific syntactic markers.

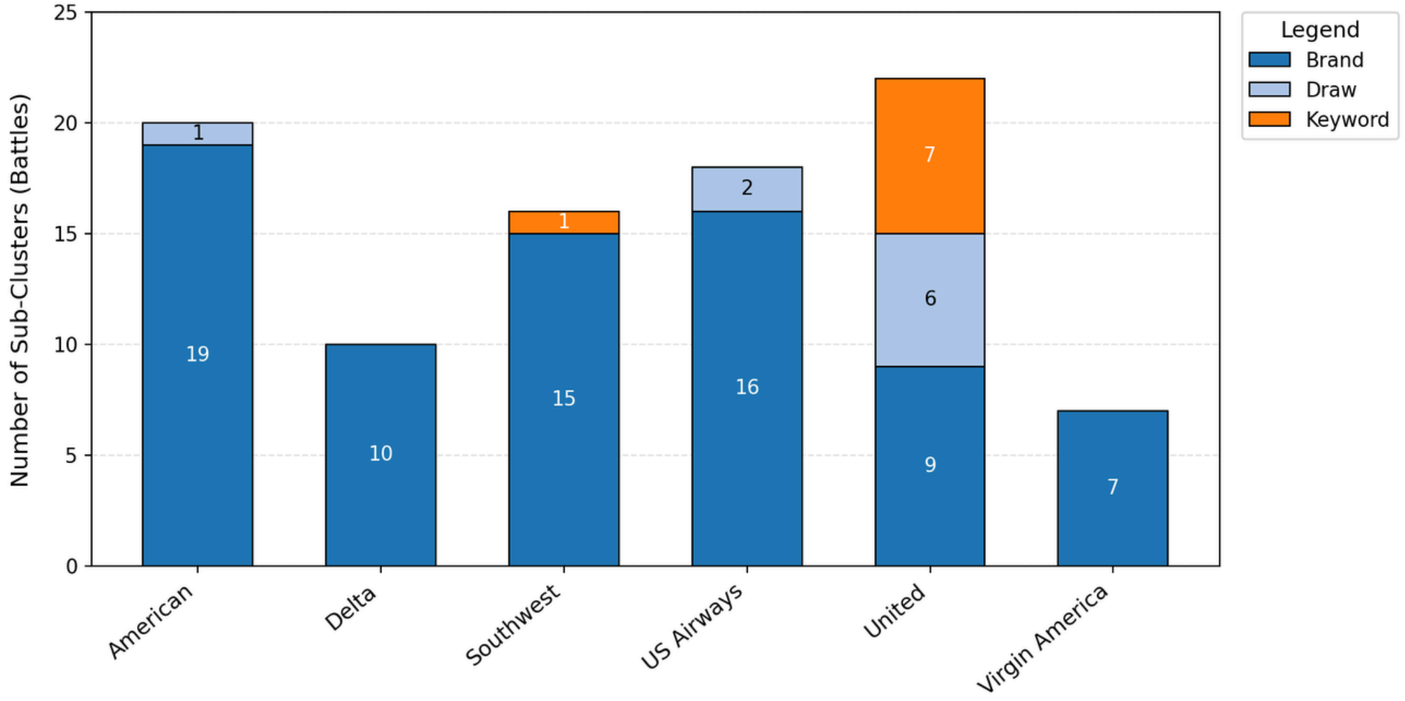
Results III: The Data-Driven Ablation

We implemented a tweet-centric ablation study across 93 valid sub-clusters (Airline + Topic) to pit the Airline Brand directly against the primary Operational Keyword.

Implementation:

- For each tweet, we performed a paired ablation by removing the airline handle and the target keyword separately, embeded the resulting variants with Qwen3, and used a Wilcoxon signed-rank test ($p < 0.05$) to identify the dominant spatial anchor across sub-clusters.

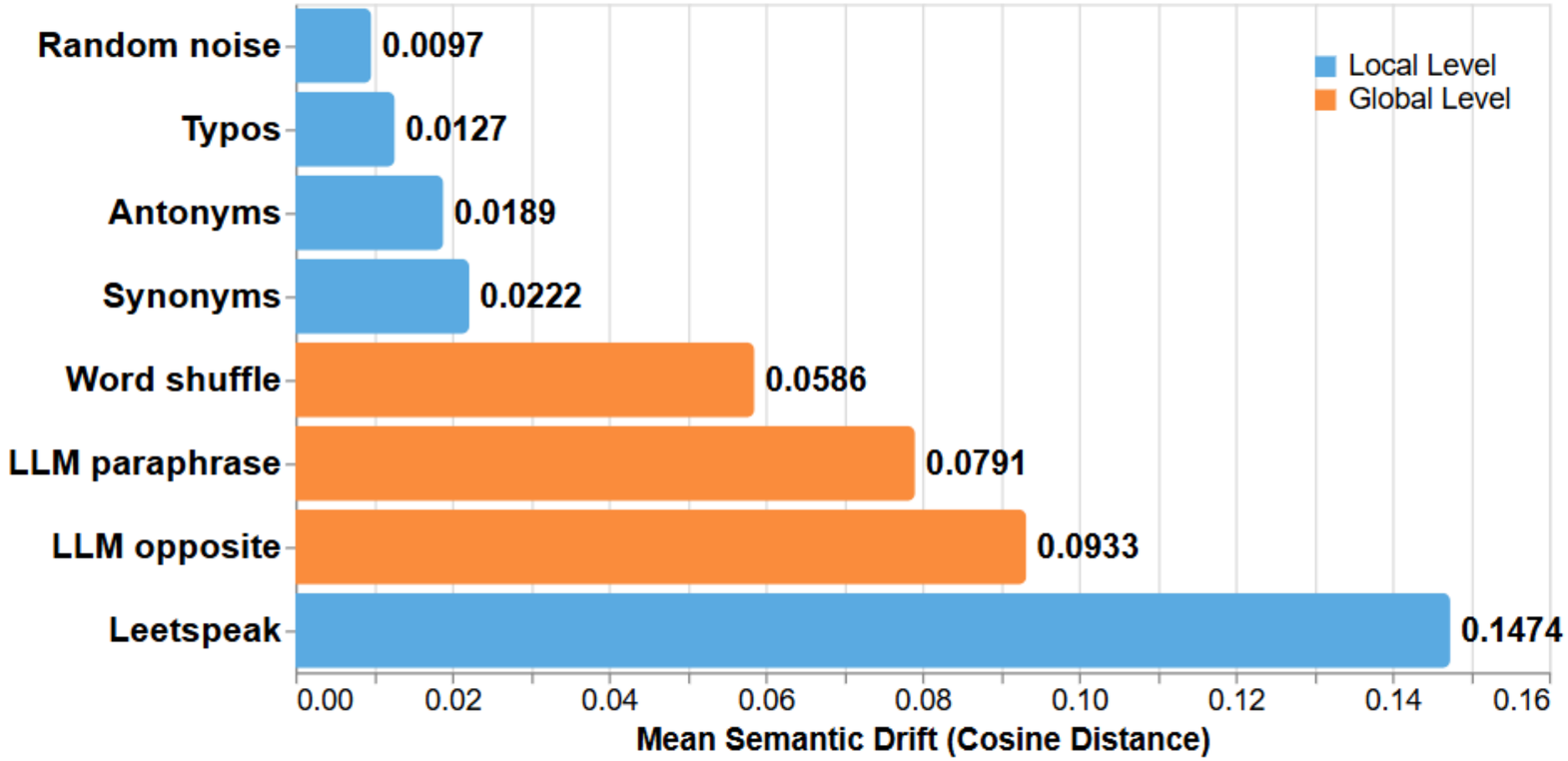
- **Peculiar Finding:** United Airlines registered the lowest semantic impact despite being the predominant company.



User-level interaction : The user can select a region, inspect its representative keywords and compare the different effects of removing the airline brand and the operational keyword.

Results IV: Robustness Analysis

- **Robustness** to small lexical and noisy perturbations. Individual word/character-level changes usually preserve the original embedding position.
- **Sensitivity** to structure and context perturbations.



User's action flow

Select a tweet → Inspect the heatmap → Apply diffrent perturbations → Observe the trajectories

Main Insight

Qwen3 is robust to small local perturbations, but more vulnerable to structural and contextual transformations.

Conclusions and Future works

- Our analysis shows that the **Qwen3** embedding space is often driven more by **syntax**, **urls**, and **airlines** than by the actual meaning of the text. While the model is robust to small lexical changes, it becomes more unstable when sentence structure or context is modified. Our visual framework helps users to reveal these hidden dependencies, prevents misleading interpretations of clusters and analyze the behavior of the embedding model

- Future work will compare different and larger embedding models to understand whether these biases are model-specific or decrease as model scale increases.



Thank You

